

BLDC Motor Controller

Design Pointers

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Bulk Capacitance

- Use as much as you can
 - A typical amount used is $\sim 10\mu\text{F}/\text{A}$ rating, e.g. VESC 75300 has $6 \times 470\mu\text{F} = 2820\mu\text{F}$.
 - The bulk capacitors protect against bus voltage ripple and the size of them is determined by the acceptable ripple. Most people would accept $\sim 1\text{-}5\%$. Ripple is of particular concern when operating near voltage ratings of parts.
 - The worst case ripple is seen at low modulation indices, and is discussed by Benjamin in good detail at <https://www.youtube.com/watch?v=x6IPdI9OVQg>
 - and by user Zombiess <https://www.youtube.com/watch?v=LcJCTB-6k2c&t=800s>
- Don't use the largest capacitance MLCC for a given package size, go down a capacitance tier.
 - Speculation/experience hints that the most stressed capacitors 1) crack and 2) suffer from capacitance drop with voltage more severely
- Place your bulk capacitors as close to the MOFETs as possible
 - Idea is to minimise the inductance between the capacitance and the MOS. Bear in mind that the electrolytic capacitors themselves have some inductance, in the order of 10s of nH so there is diminishing return on this.
 - Ceramic capacitors have much lower inductance, and ESR, so should be placed as close as possible to the MOS.
- The amount of bulk capacitance that is needed is based on the motor current, switching frequency, battery/wire/motor inductance. Strictly, if the battery and wires had no inductance, then no capacitance would be needed, but this is unrealistic.
 - Estimating it is very tricky, it depends on many things, and it's one of those ones where simulation does not seem to be easy to get representative answers.
 - Vesc 100250 has bizarrely low capacitance.
- Remember MLCC capacitors reduce in capacitance when they have a voltage across them. Graphs of this phenomenon are posted in the Gate Drivers section.

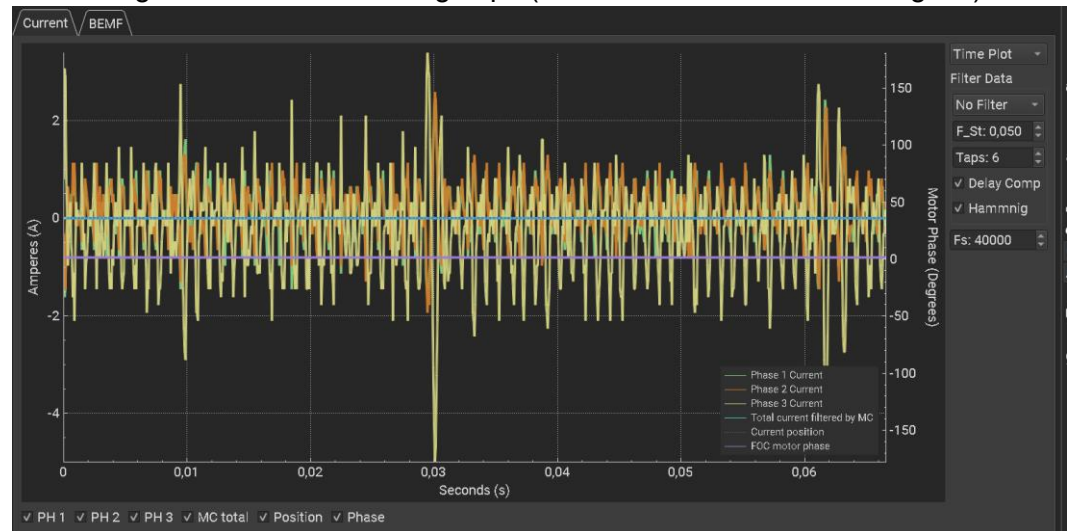
Voltage Regulation

- Generally:
 - use a low ripple 5V supply (current mode regulator)
 - use a linear regulator with high ripple rejection. ($>60\text{ dB}$)
 - have at least $100\mu\text{F}$ on 3.3V rail
 - have a maximum of 5-10mV ripple on the analog reference input. If you filter that input, recommend at least $22\mu\text{F}$ post filter.

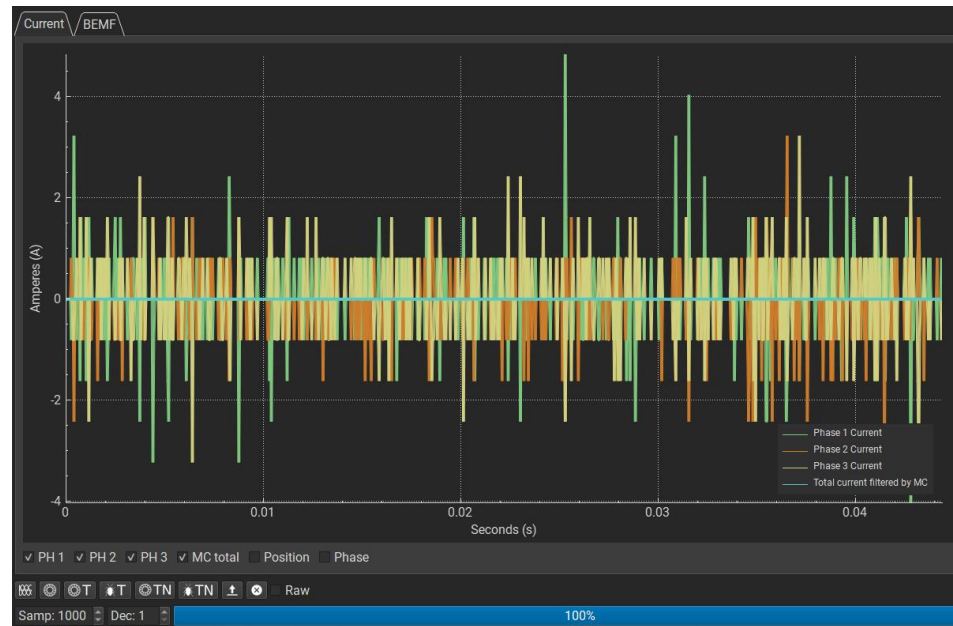
- Wrong (note the $\pm 5A$ noise on the estimated current):



- After fixing the above with filtering caps (noise reduces to 20% the original):



■ Ideal:



- **Mid-High voltage:** If your ESC is rated for 60V choose a much higher-rated switch mode regulator chip. They are prone to very unconservative ratings, and are frequently the first thing to die in a VESC crash. @mxlemming recommends a 100% margin with TI regulators, whereas LTC regulators seem to require much less margin, perhaps only 15%.

Preferred Voltage Regulators

This is a non-exhaustive list, just some that are popular.

Expensive and good:

LTC7103 or LTC7101 (100V)

LTC7138 (140V)

Many dc/dc bricks e.g. CUI pqde6w-q110 and many pin compatible devices

Popular and terrible:

LM516x from TI (100V)

LM5017 from TI (100V)

Voltage Sense

- If using non-isolated voltage sensing, consider the power loss to the resistors. Typically the high side resistor is the one that will be critical.
- Surface mount resistors do have a voltage rating.
 - 0402 → 50V
 - 0603 → 75V
 - 0805 → 150V
 - 1206 → 200V
 - 2512 → 500V
- Target the MOS max voltage as $\sim \frac{2}{3}$ to $\frac{3}{4}$ of the ADC reading, to enable some headroom for voltage spikes. So for example, 84V FETs being read by a 3.3V ADC should result in $3.3 \times \frac{2}{3} = 2.2V$ on the voltage divider; so a potential divider of about 40 is needed.

You may choose for example 80k Ω and 2.2k Ω to achieve approximately this ratio, and the power loss will be $84V^2 \div 82000\Omega = 0.086W$

- A similar divider can be used for the phase voltage sensing.

Current Sense

- 3 current sensors are superior to 2 current sensors due to the ability to read the currents accurately at high modulation.
- Phase current sensors are preferable to low side, since the recovery time after switching is typically lower, and the possibility of filtering is better.
 - However, above a certain voltage (typically 16s battery) there is no choice but to use low side current sensors, or isolated sensors. This is because of the difficulty finding high-voltage compatible current sensors.
- Low side sensors are much easier to implement high voltage systems with, BUT the utility of these is limited since the safety aspect starts to become important above the typical limit of the INA240 typically used by 16s/75Vmax controllers.
- **!!!! IF YOU USE LOW SIDE SHUNTS, DO NOT CONNECT CAPACITORS TO THE LOW SIDE MOSFET GROUND, THEY MUST GO TO THE GROUND OF THE SHUNT !!!!!**
- Current Trampa VESCs use:
 - VESC 4.x uses low side 2x shunts
 - VESC 6 uses 3x phase shunts
 - VESC 75/300 uses 3x phase shunts with INA240
 - VESC 100/250 uses 3x low side shunts with...?

Preferred Current Amplifiers

This is a non-exhaustive list, just some that are popular. Use the lowest gain version as they normally have a higher bandwidth.

- INA240A1
- INA181A1
- AMC130x (isolated, high voltage)

Alternatively, it is perfectly acceptable with low side shunts to use a standard opamp wired for differential reading of the shunt. Typically however, this arrangement will have a more limited common mode input range, so take care with amplifier selection (must be rail to rail in (and out) and the design must not allow voltage spikes or readings to the opamp inputs below ground). You will need 5MHz+ GBW, since you have a requirement for ~20 gain on most motor drives, giving a 250MHz response speed.

Good choices of parts are:

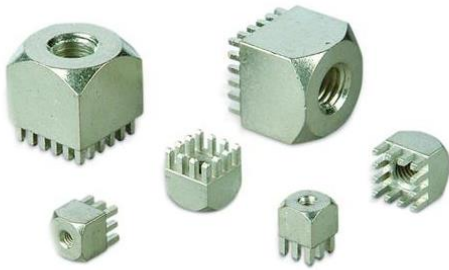
- NCS2003x
- COS72x

Some parts that look suitable may not be due to input range and the possibility of phase inversion.

Connectivity

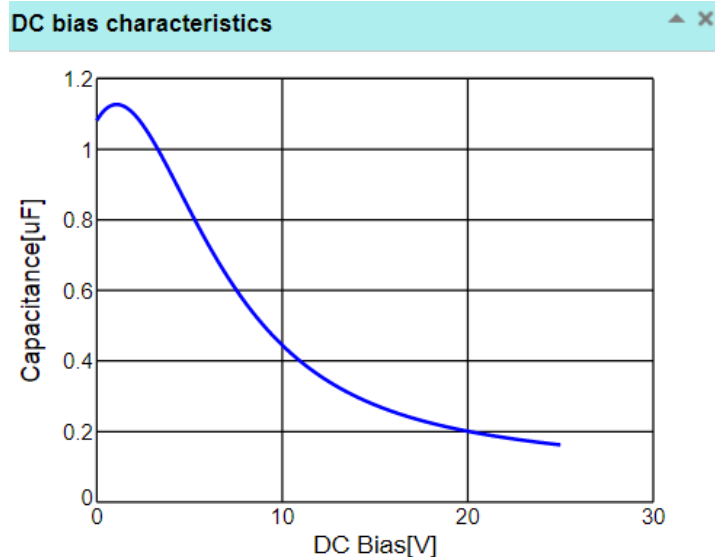
- Use latching connectors if you can
- Include protection against static discharge and shorts

- Solder pads for high current connections require large soldering irons (>50W) consider using power element screw terminals.

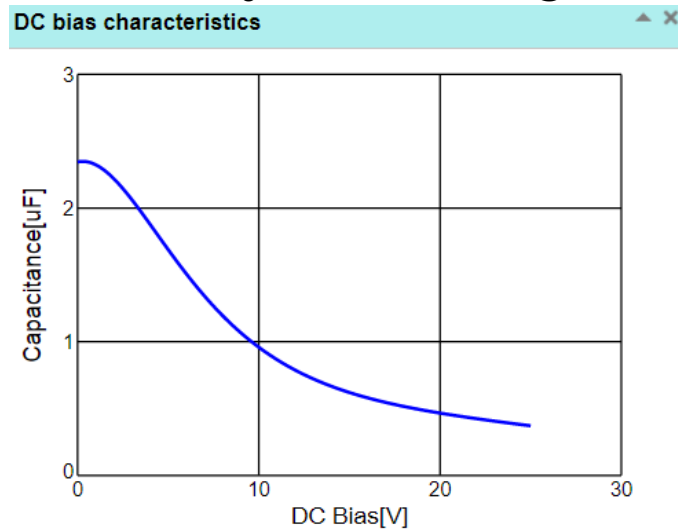


Gate Drivers

- [UCC27211](#)'s seem nice on paper, but have no input filtering and have been shown to falsely trigger
- Bootstrap capacitors need to be at least 10x larger than the total gate capacitance
- Mosfet gate charge is specified in nC, so to work out what size capacitor to use for the bootstrap (capacitor floated using a diode to supply voltage for driving the high side mosfet), take the nC given and divide by the voltage to get to nF. Then times by the number of mosfets in parallel and times the result by 10x to get a safe value.
 E.g. $200\text{nC} @ 10\text{V} = 20\text{nF}$ 3 mosfets in parallel so $20\text{nF} * 3 * 10 = 600\text{nF}$.
 However ceramic capacitors need to be derated when a voltage is applied.
 A typical $1\mu\text{F}$ 25v 0603 capacitor will derate to approx 350nF at 12V.
<https://ds.murata.co.jp/simsurfing/mlcc.html?lcid=en-us>



So in this case to get the wanted 600nF @ 12V, a 2.2uF 25V 0603 would need to be used.



Ceramic capacitor values can vary more than 50% with temperature, pay attention that capacitance is sufficient at the worst case temperatures.

Preferred Gate Drivers

This is a non-exhaustive list, just some that are popular.

Simple - MOSFET drivers

- Those weird 2edxxxx ones
 - 2EDL23N06 (pseudo isolated, with overcurrent sense for low side shunts and UVLO feedback) - Successfully used by mxlemming (note issues with built in bootstrap diode resistance, not much good at higher frequency and with big MOS)
 - [2EDF7275KXUMA2](#) (isolated with UVLO feedback) Successfully used by Fajdiga, mxlemming and Tchaumnu. These things have very very strong drive capability.
- FD6288Q - it just works, high voltage rating, cheap and decent output current, 200ns deadtime. Successfully used by Tchaumnu
- NCV5183 NCP5183 (high current (4A) basic, very robust variant of the IR2181). Default cheap driver of mxlemming. There are many pin compatible equivalents from Diodes inc, Fairchild, TF Semi...

- UCC21520
 - Isolated gate driver for high voltage and or large physical size inverters, they allow large spacing between the phases, and strongly avoid parasitic turn on of the inputs. Also useful at high voltages that would be harmful if touched.
 - Analogues of this include Onsemi parts NCP51561

Complicated - IGBT drivers

Avago, Broadcom, Infineon, ST... consider desaturation, miller clamps, slower switching, different speeds at charge and discharge by using diodes...

- NCV57000

Notes on Gate Driver Usage

- 2.2uF+ ceramic capacitance close to the low side of each driver. This supplies the low side drive and the bootstrap
- Bootstrap should be ~10-30x the MOSFET gate capacitance depending on the derating of the capacitor.
- If tying the gates low with resistance, consider the rate at which this will discharge the bootstrap (typically gates should be tied low with 100kohm+, it is only to stop them floating high if unpowered/gate drives unplugged)
- Gate driver current ratings are vague, but typically means it is the max current they will supply if driving a short circuit. Consider this when calculating the gate resistance. You do not usually need to add a resistor that would limit the gate driver current to below the rating. A 1A gate driver will drive 1A peak into a MOSFET with no resistor. Adding a 12 ohm resistor at 12V will halve this to about 0.5A.

Gate Resistors

$$I_{Drive} = \frac{Q_{gd}}{t_{RiseFall}} =$$

$$R = \frac{V_{GS}}{I_{Drive}} =$$

$$P_{Norm} = I_{Drive}^2 * R$$

- Power calculate is peak power when the gate driver is driving current into/out of the MOSFET gate
- "Just assume all the energy put into the gate gets absorbed by the gate resistor then $1/2CV^2*f$ "

MOSFETs

- Derate MOSFETs current by 30-50%, a 300A MOSFET on paper is really a 90A to 150A MOSFET in practice. A good spec to look at is the continuous rating for a 1x1inch pad for a realistic current you can easily achieve.
- Derate voltage by ~20% if you have good control of your switching. Typically, 20s is safe with 100V MOS, 12s with 60V MOS, 30s with 150V mos. There is no hard and fast rule, erode margins as you want.
- MOSFETs have a gate resistance too, remember to also include the gate driver output resistance when calculating what gate resistors to use! Start by sizing the gate resistors to have the rated current output from your gate drivers.
- MOSFET specs to look out for
 - Rds on
 - This determines the resistive losses to current
 - Vds
 - This determines the voltage rating (obvs...) Different manufacturers have different opinions on what this rating means, from "10s of uA" breakdown to "destruction at this voltage"
 - there is usually a ~13V margin to destructive collapse on a 100V MOS at room temperature.
 - Ciss
 - This is one factor that determines the rate at which the gate driver can turn the FET gate on. Typically 3-13nF for high power MOS, it is related to the gate area and thickness of the silicon in the junction between the gate mesh and the source.
 - Crss
 - The "Reverse transfer" or miller capacitance, this determines the MOS' tendency to self turn on as the switch node rises (low side FET) or falls (high side FET)
 - Modern, decent MOS have a Crss to Ciss ratio of greater than 1:100. Typically, a mosfet with $C_{iss}/C_{rss} > V_{bus}$ shows no tendency towards self turn on.
 - MOS with lower ratio (typical examples being the tediously still popular IRFP4110 with 9600pF Ciss and 250Crss, for a ratio of about 38) have a strong tendency towards ringing, as the rising and falling nodes have a negative feedback, leading to oscillation. These MOS require much more careful management, including snubbing, careful management of the drive resistance and extra gate capacitance (improving the Ciss/Crss ratio). As in the case of the VESC 75/300, it is possible to just hammer this kind of MOS on so hard that you bypass these issues, but that requires very very low inductance of the power stage to avoid ringing.
 - Avoid MOS with high Crss for an easy life.

Thermal considerations

There are both I^2R and switching losses.

I^2R losses

The resistive heat losses are significant. Under high load, MOSFETs can self-heat to the point of liquifying solder.

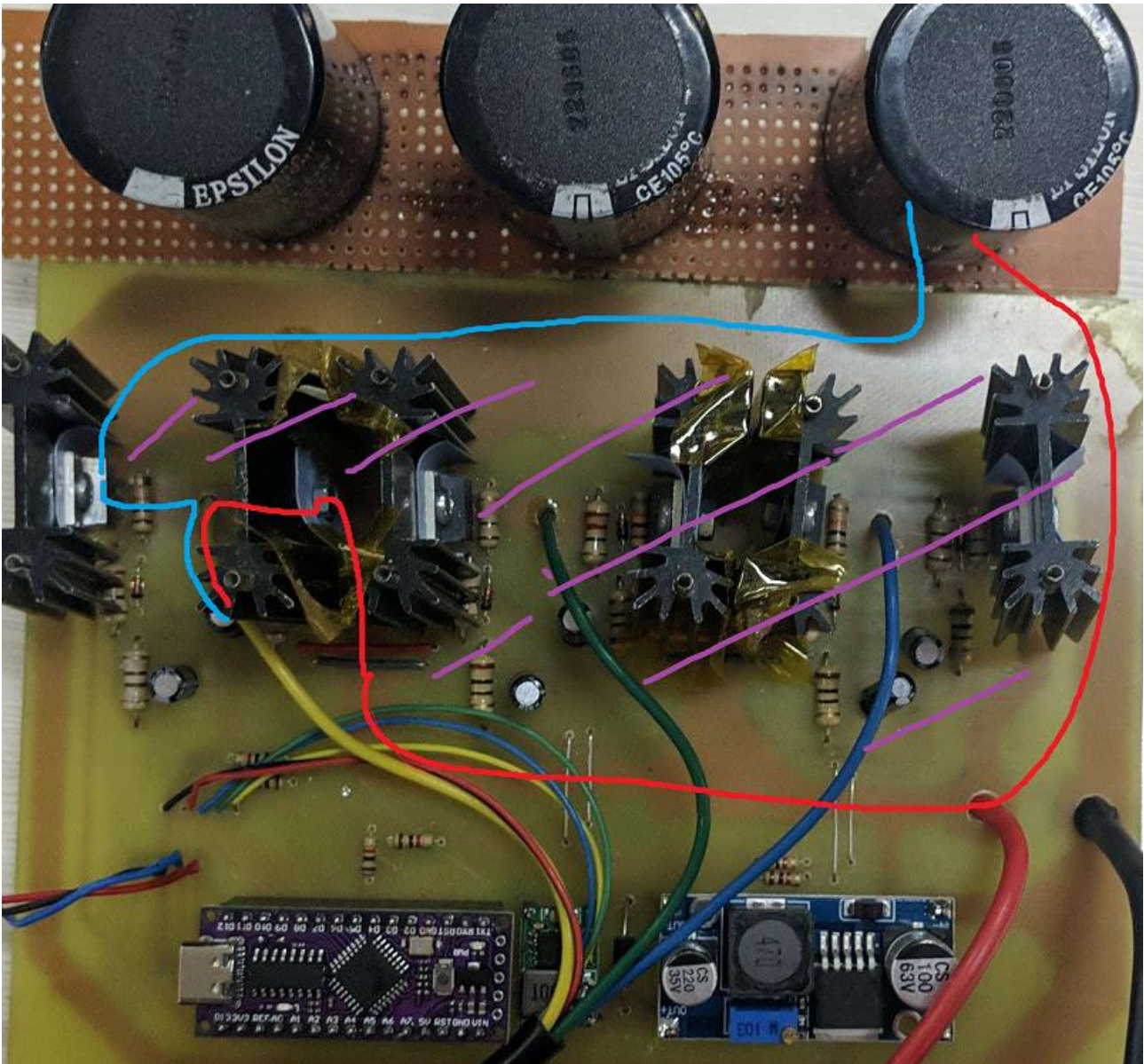
$$P_{I^2R} = I^2 * R_{ds(on)}$$

$R_{ds(on)}$ is not a constant, and varies with temperature as well as gate voltage. Thus, precise modelling and testing is important for determining maximum peak and sustained current.

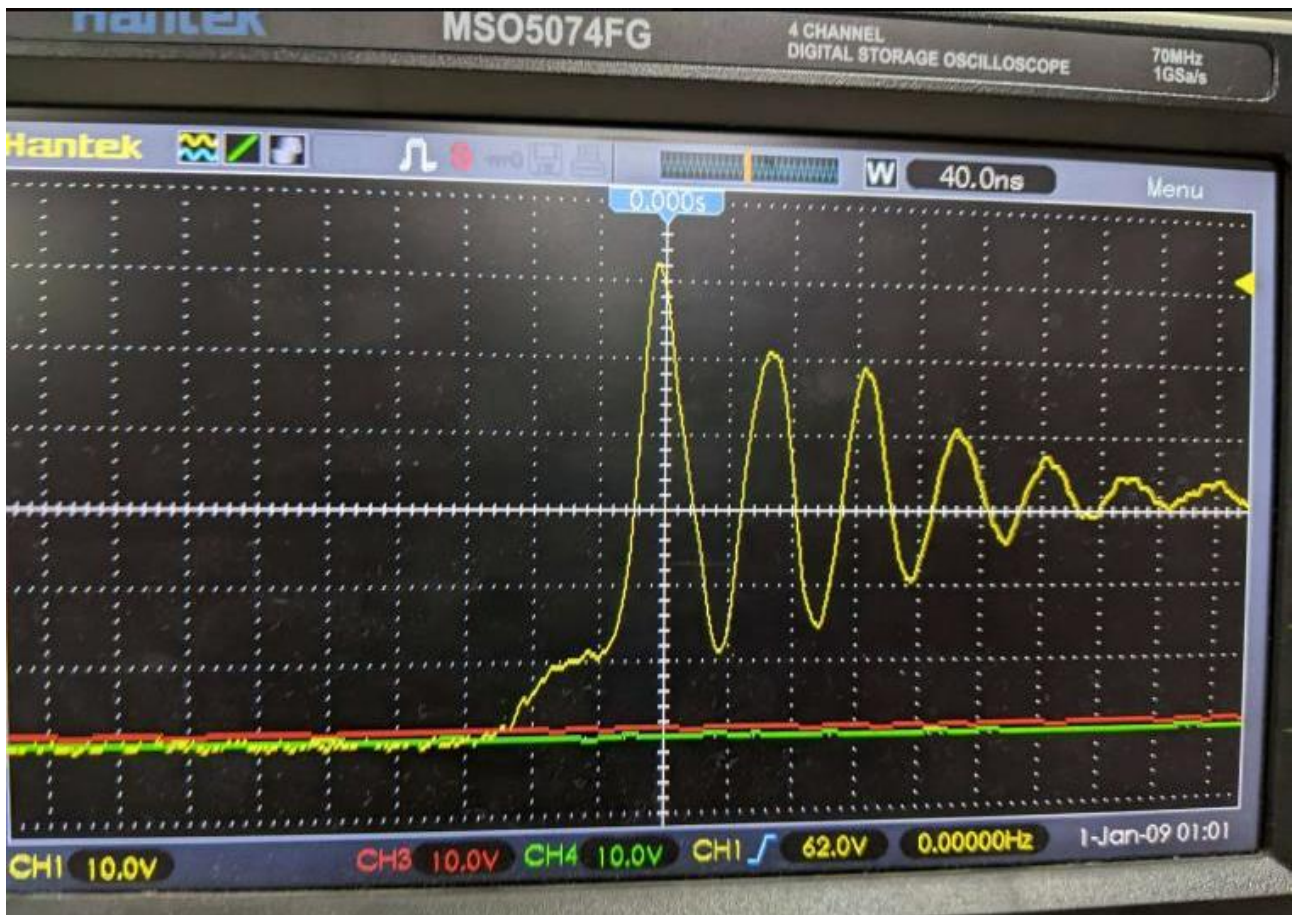
Layout Guidance

- Place the current amplifiers as close to the microcontroller as possible. The traces from the current shunt should be routed as a differential pair and connected at the shunt with a kelvin connection. The differential pair to the amplifier is much more immune to noise than single ended (single line) signal to the MCU, so keep the amplifier close to the MCU and route the track on an internal layer with a good return path (unbroken ground path under the trace)
- Gate drivers are also better to be in a place where the signal lines from the MCU are not near any high dv/dt traces as these lines are sensitive to noise. The traces to the gates of the mosfets should have parallel return paths on a layer directly above or below. (or routed as differential pairs if you have isolated drivers).
- **DC link inductance is the single most important thing to optimise as it directly affects switching ripple.** Keep paths to ceramic capacitors as short and thick as possible and remember the TOTAL distance matters, including the ground return, not just the distance between the capacitor and the high side of the mosfet.
- The DC link between the H-bridge legs is very important. Do not rely on battery leads to make the link. The full current is switched between the high and low side at the switching frequency (typ. 15kHz)
 - It is possible to create separate boards for each phase, but care must be taken to make ultra low inductance links between the boards, and it may become essential to use isolated drivers and take special precautions with current shunts.
- Don't place vias on pads if at all possible. It can cause the solder to wick down the vias and there will be no solder on your parts. Most SMD parts are designed to float on the surface tension of the molten solder and get pulled into the correct alignment, so if all the solder disappears down the vias the parts might end up in the wrong place and short out or have a bad connection / uneven height. This is particularly important when mounting a heatsink over an array of SMT mosfets as the uneven heights will create hotspots.
- Online calculator for modelling trace inductance: <https://spok.ca/index.php/resources/tools/106-traceindcalc>
- Offline tool for inductance and capacitance and more: <https://saturnpcb.com/saturn-pcb-toolkit/>

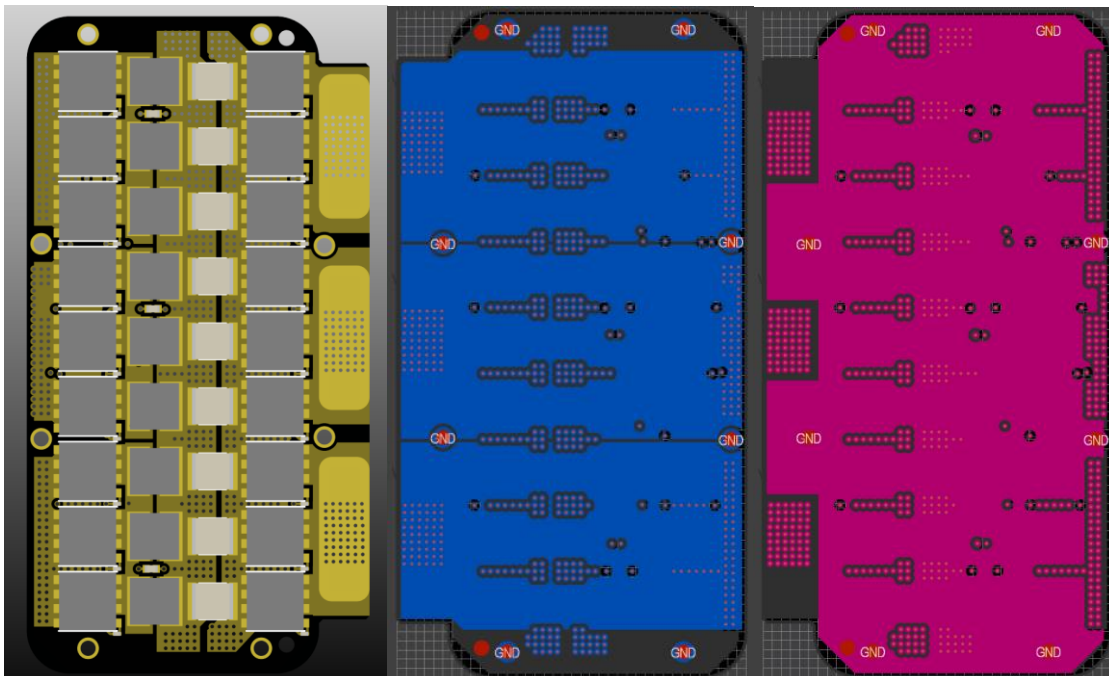
- PCB trace resistance is a major source of heat. However, thicker pours make soldering components more challenging.
 - A 4oz pour is the reasonable upper limit for soldering FR4-backed PCB.
 - Members have reported that 3oz pours of Aluclad are impossible to solder.



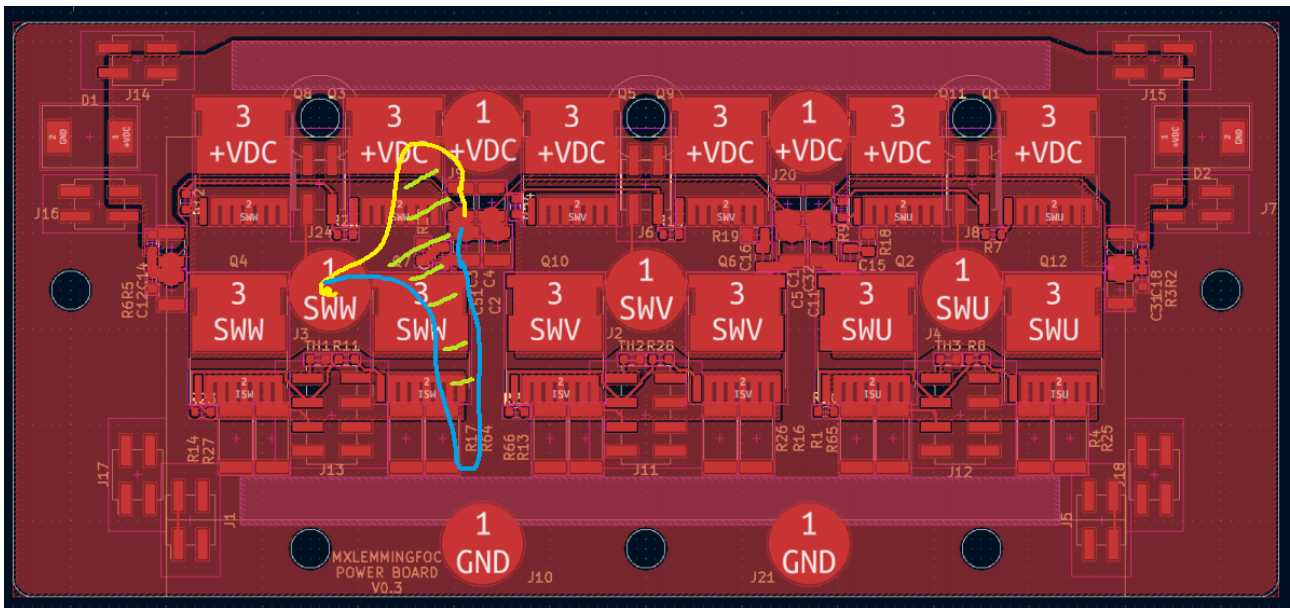
An example of a large switching loop



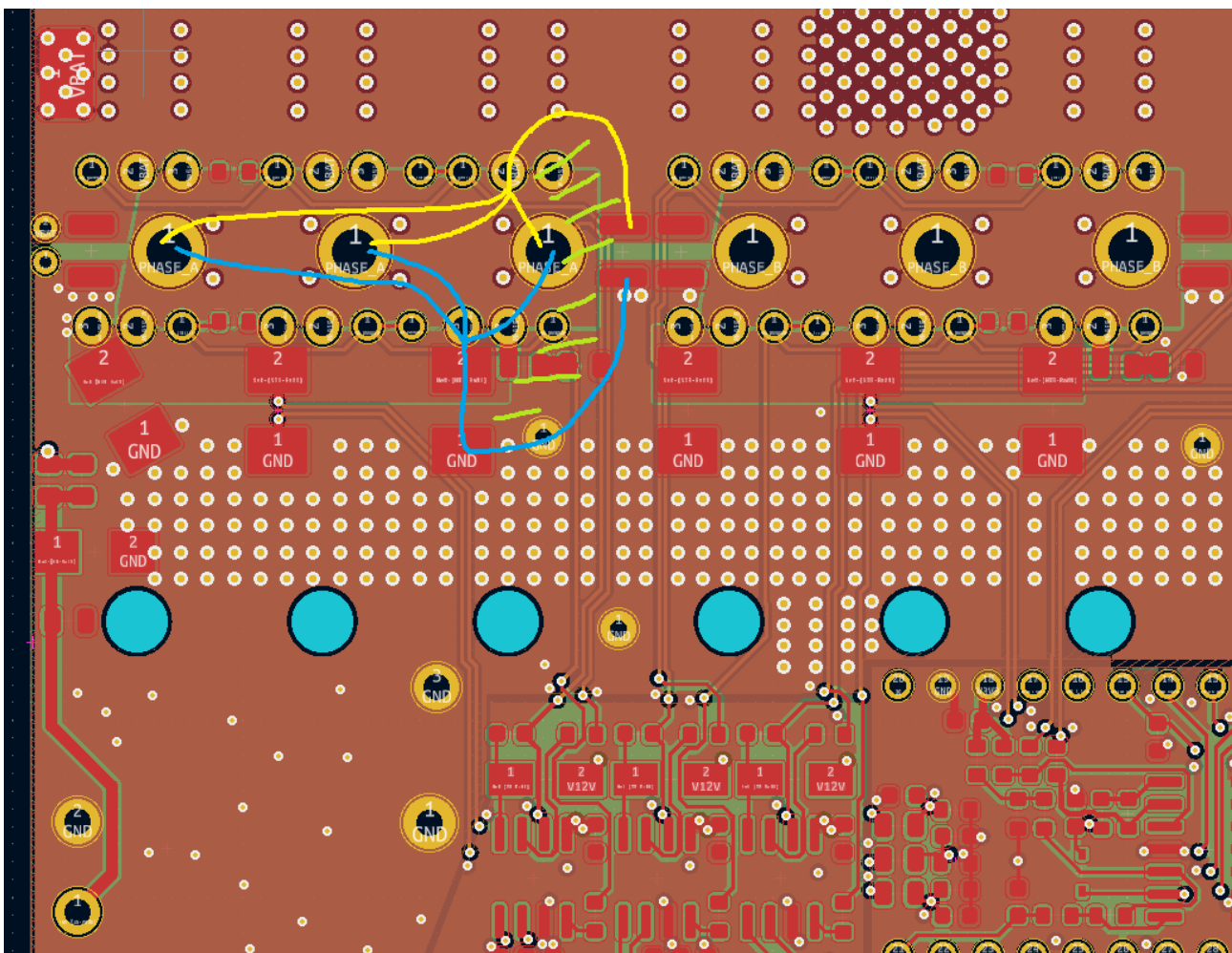
Such a large switching loop results in this on the switch node



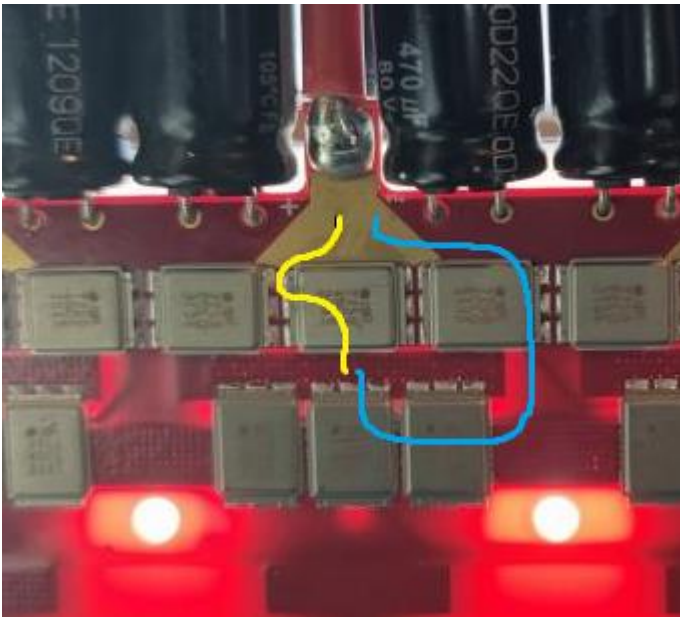
A layout with very low parasitic inductance and a tiny switching loop (loop is within the board FR4) (Techaumnu A200s)



A single layer board with very low inductance (mxlemming IMS)



A through hole stage with very low inductance (Badgineer MP2/CCC ESC). Note the continuous ground plane underneath everything; the only significant inductance is the legs of the MOSFETs themselves.

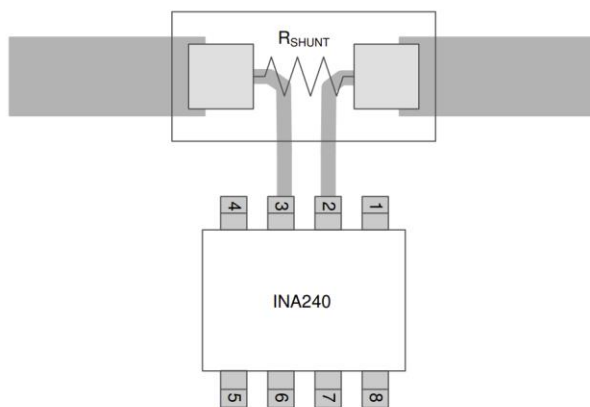


The VESC 75300 makes the slightly odd choice not to have a ground plane, but the switching loop is still tight and the traces wide so can carry 480A+ for short periods without failure.

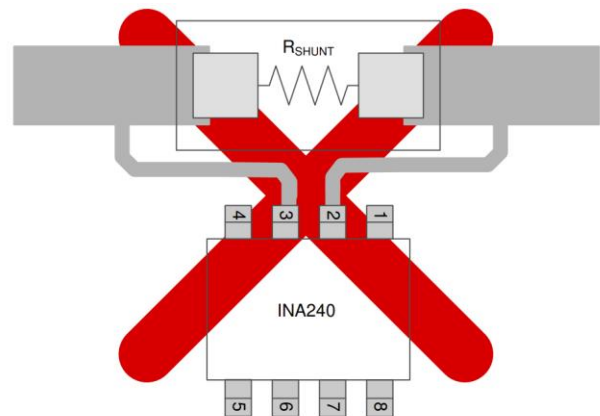
Kelvin connections for high-current sensing

At the high currents where the shunt resistor possibly has *lower* resistance than the PCB, it is extremely important that all Kelvin connections be properly made. Pay special attention that any vias used do not accidentally connect the shunt sense wires to current-carrying planes.

For more information about good Kelvin trace layouts, see: <https://www.analog.com/en/analog-dialogue/articles/optimize-high-current-sensing-accuracy.html>



DO
Kelvin Connection from Shunt Resistor



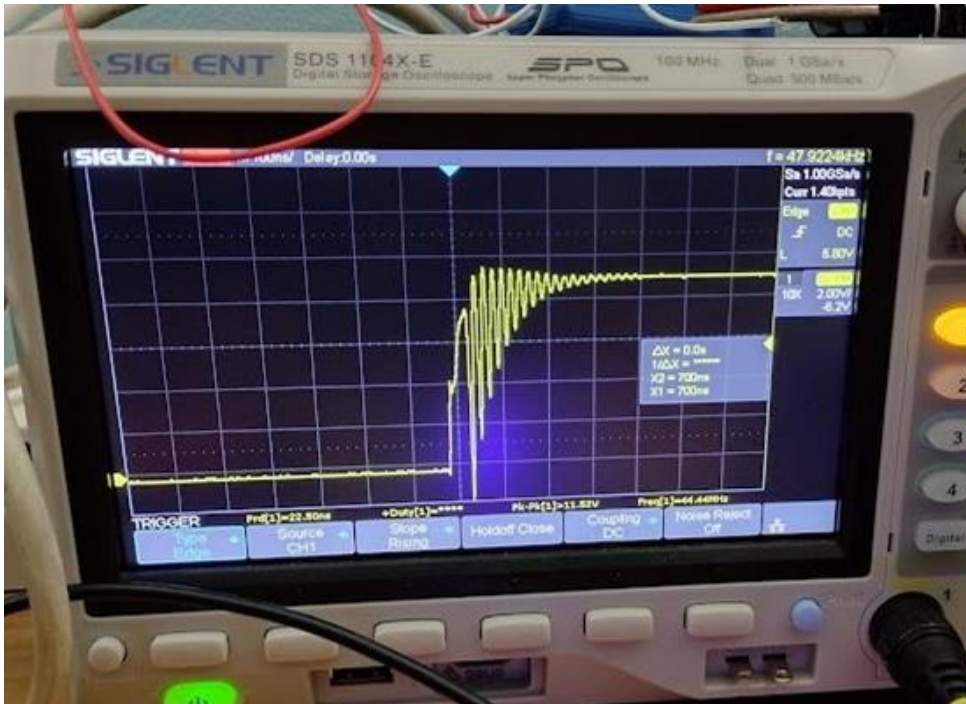
DON'T
Non-Kelvin Connection from Shunt Resistor

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Figure 9-7. Shunt Connections to the INA240

Troubleshooting

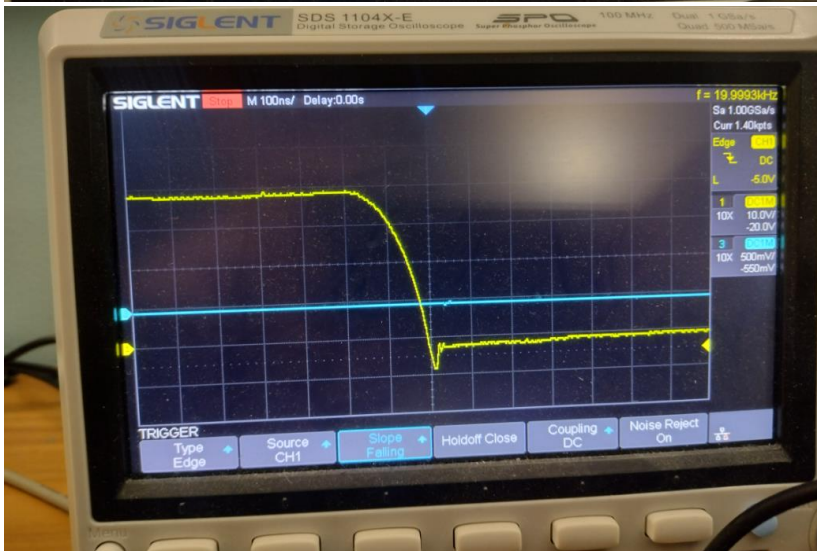
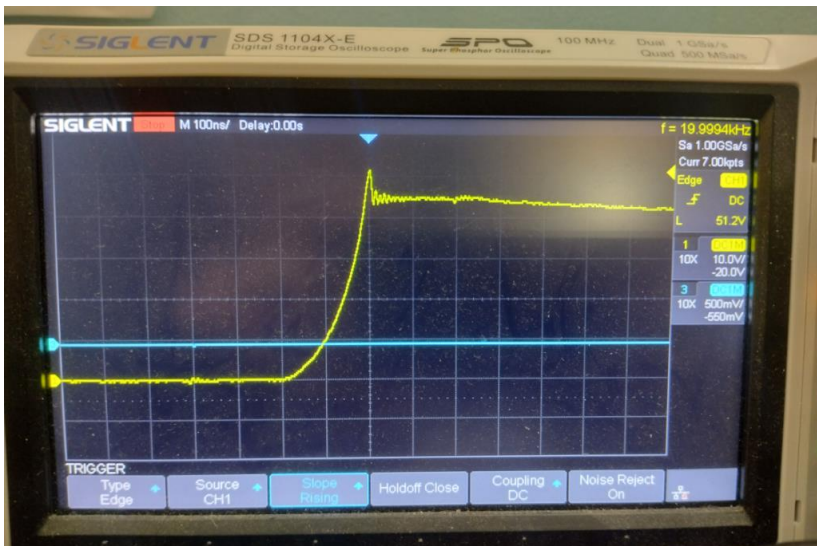
Ringing



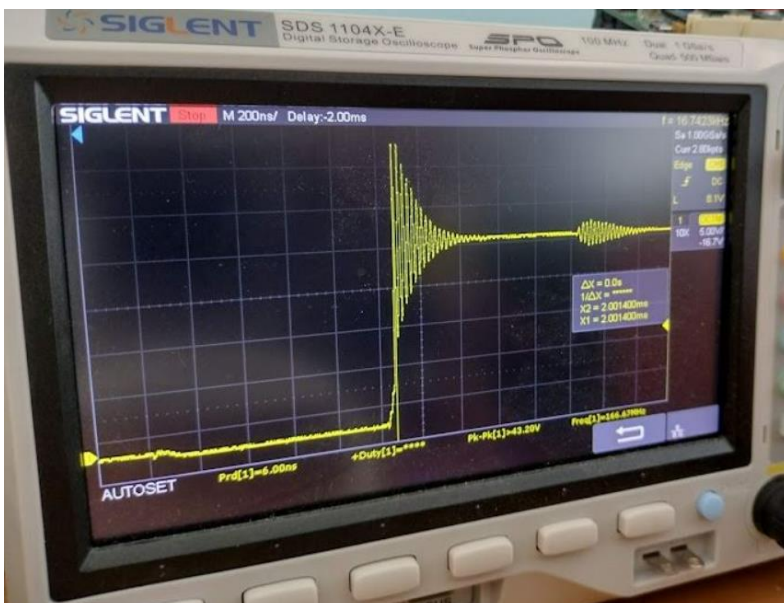
Bad ringing on VGS

Ringing can cause significant issues in the hardware, up to and including causing the FETs to overheat and fail.

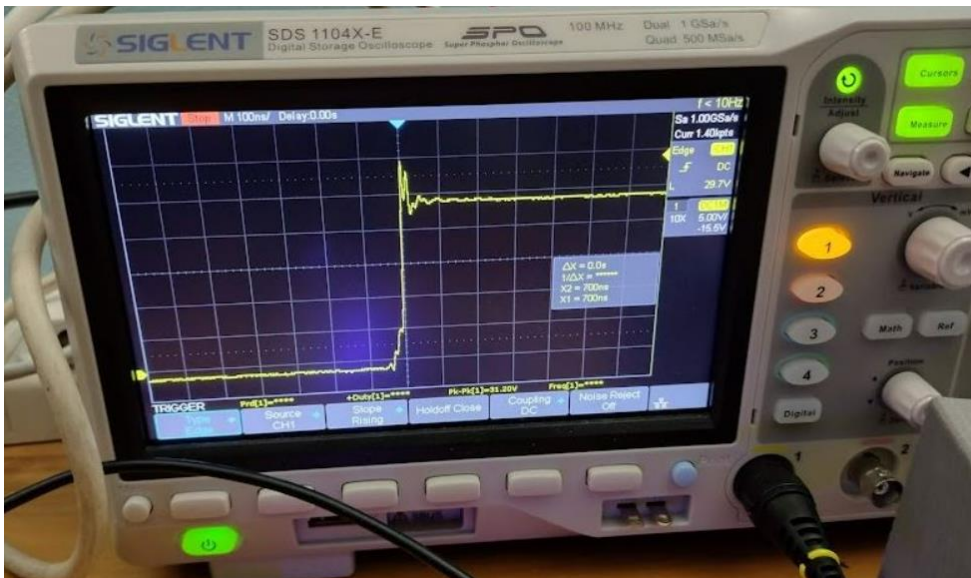
<https://endless-sphere.com/forums/viewtopic.php?f=30&t=113108&hilit=snobber&start=75#p1694327> details an approach to resolving ringing issues.



Nice switching waveform at 200A. TOLL with low Crss on IMS board



Not nice waveform at only 10A - through hole TO-220 MOS with high Crss



The same through hole stage after adding optimised snubbers

Gate-drive current-limiting resistors

Higher resistor-values slow down the gate turn-on, which in turn reduces high frequencies.

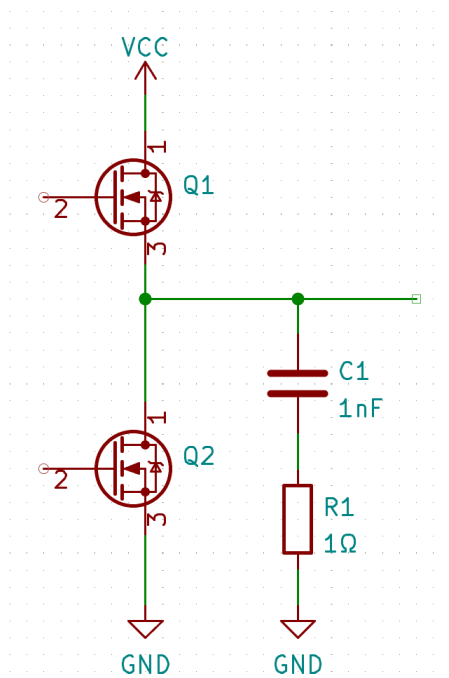
Good values result in gate time constants between 100 and 300ns for silicon MOSFETs. This usually means a total gate resistance per MOSFET of 4.7 ohm to 30 ohm for mid size inverters.

GaN should be 5-10x faster, and IGBTs 2x slower, typically.

Snubbers

Sometimes necessary for ringing management. Snubbers should be calculated and fitted after the hardware design is done. Allow footprints if your design is likely to ring (through-hole MOSFETs are particularly susceptible).

The snubber will typically be in the range of 1-10nF– *nano*, not *micro*, unless you like fire!– and 0.5-3 ohm resistance. The exact size is calculated by noting the ringing frequency, the switch node capacitance (basically the MOS Coss), and therefore calculating the damping factor for critical damping. The RC frequency should be similar to the ringing frequency (after accounting for the change in frequency due to the RC) as this results in the maximum power transfer to the snubber.



Many of us use:

<http://paulorenato.com/index.php/electronics-diy/197-rc-snubber-calculator-spreadsheet>

To calculate values.

Snubbers are only needed from switch node to ground. For high frequencies, the Vbus and ground are one and the same, since they are coupled to each other through the ceramic capacitors and the capacitance between planes.

Typically, snubbers will need 0805 components for <60V operation and 1206 for <100V operation. Your mileage may vary, depending on your MOS choice and ringing found.

The capacitor determines the power dissipation, evaluated as

$$P = \frac{1}{2} \times C \times V^2 \times F_{zv}$$

The resistor dissipates the energy.

The components should be rated for the full bus voltage. These components have a hard life, overrate them.

Miscellaneous Suggestions

- Use a current limited power supply for initial testing, start at the lowest voltage and current and gradually increase the allowed current until you know it is working ok. Then you can start increasing the voltage.
 - If you stay below 15V, on your “Vbattery” line, it is quite hard to kill the gate drivers.
 - If you current limit at about 200mA, it is quite hard to kill anything. This is enough at 12V to power up your whole board, program the MCU, check all the voltages...
 - If you start at 3V3 and gradually wind up while watching the 3V3 and 5V and 12V lines, you stand a very good chance of finding errors and correcting them before you cremate your... everything small.
- Don't stick your face over the electrolytic capacitors, they make a right mess if you get them the wrong way round!
- For testing gate drivers is a diode test from gate to source and gate to 12V/bootstrap. You will see the protection diodes one way and an open circuit the other way.

Double Pulse Testing

- 1) gradually increase the voltage and time on for the first pulse looking out for issues
- 2) Try to start out operating in a point where the energy in the cap bank is smaller than the rated avalanche energy of the FETs. It is quite hard to kill things if there is simply not enough energy available to kill it.
- 3) make sure you have everything all neat, tidy, board free of flux and solder bridges before you start - my experience on DP testing to date has been that you can push eyewatering currents without any issue at all... On the assumption that your board has no issues that would have likely caused death in normal low power use."